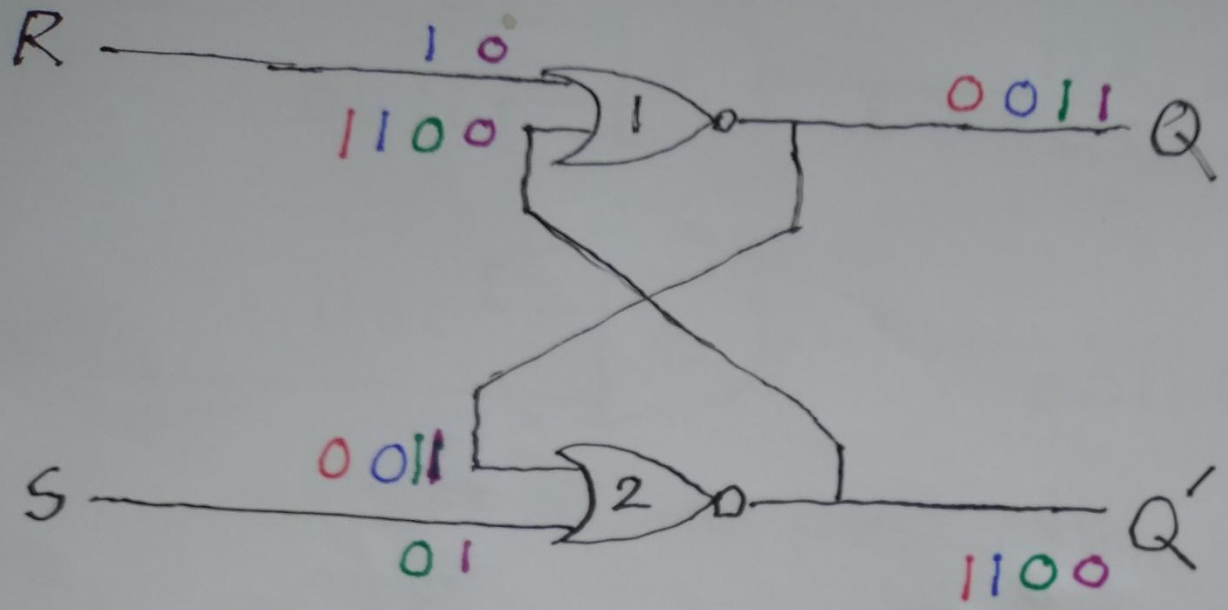
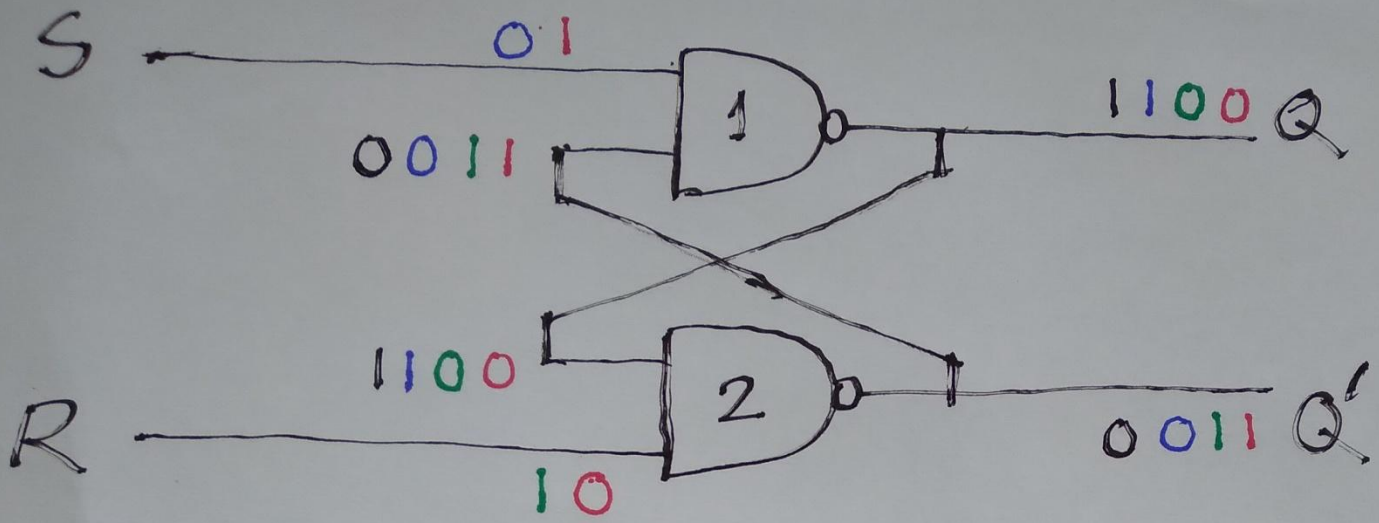


Basic FF ckt with NOR gate

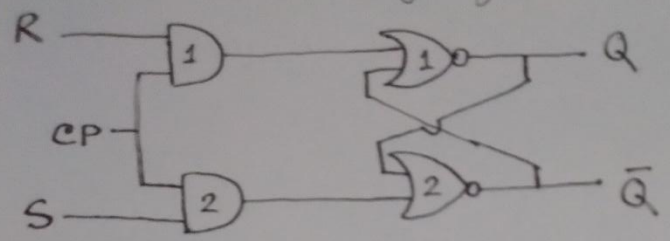
S	R	Q	Q'
1	0	1	0
0	0	1	0
0	1	0	1
0	0	0	1
1	1	0	0



Basic FF ckt with NAND gate

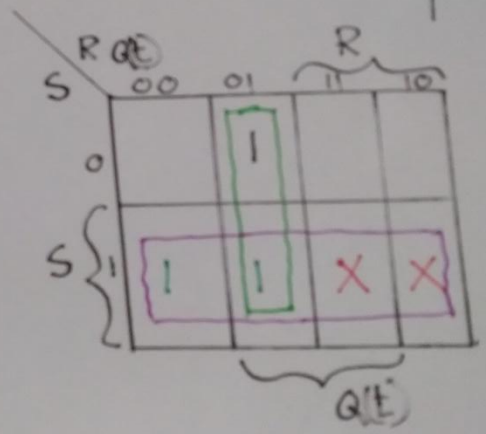
S	R	Q	Q'
1	0	0	1
1	1	0	1
0	1	1	0
1	1	1	0
0	0	1	1

Clocked RS flip-flop



S	R	Q(t)	Q(t+1)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	Ind.
1	1	1	Ind.

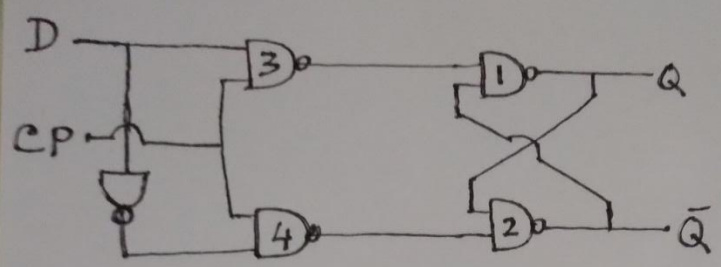
S	R	Q(t+1)
0	0	Q(t), No change
0	1	0 , Reset
1	0	1 , Set
1	1	Ind.



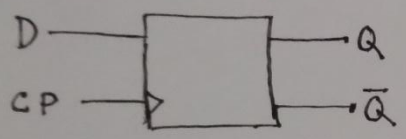
Q(t)	Q(t+1)	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

$$Q(t+1) = \bar{R}\,Q(t) + S$$

DFlip-Flop



D	Q(t)	Q(t+1)
0	0	0
0	1	0
1	0	1
1	1	1



D	Q(t+1)
0	0
1	1

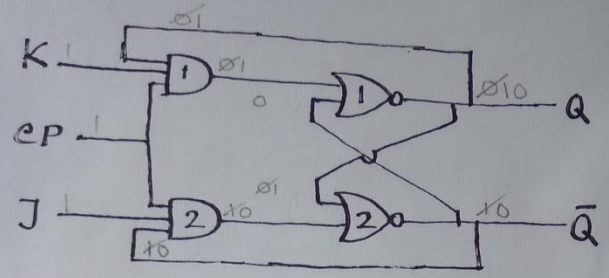
D \ Q(t)	0	1
0	0	0
1	1	1

$Q(t+1) = D$

JK Flip-Flop

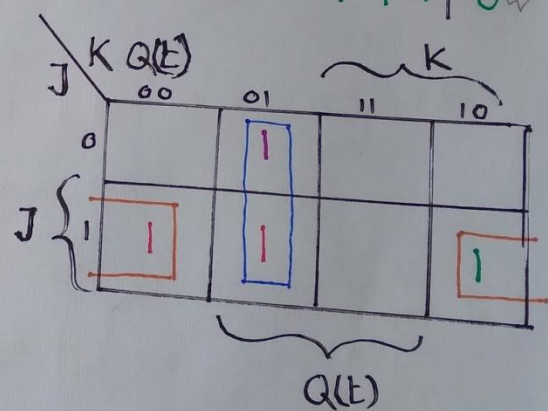
X	Y	X+Y	$\overline{X+Y}$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

NOR
Both input 0 i.e, output 1
Any " 1 i.e, " 0



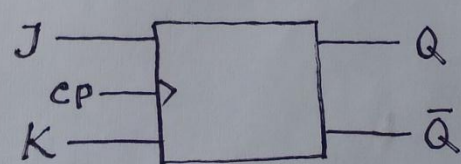
J	K	Q(t)	Q(t+1)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

J	K	Q(t+1)
0	0	Q(t), No change
0	1	0
1	0	1
1	1	$\overline{Q(t)}$

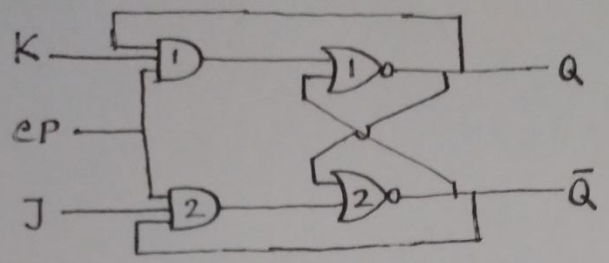


Q(t)	Q(t+1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

$$Q(t+1) = \bar{K}\,Q(t) + J\,\overline{Q(t)}$$

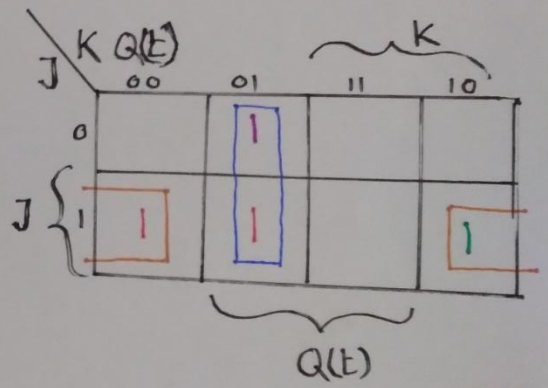


JK Flip-Flop



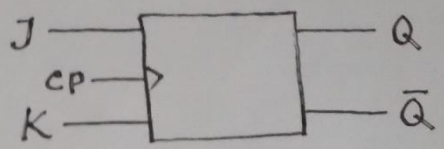
J	K	Q(t)	Q(t+1)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

J	K	Q(t+1)
0	0	Q(t), No change
0	1	0,
1	0	1
1	1	$\overline{Q(t)}$

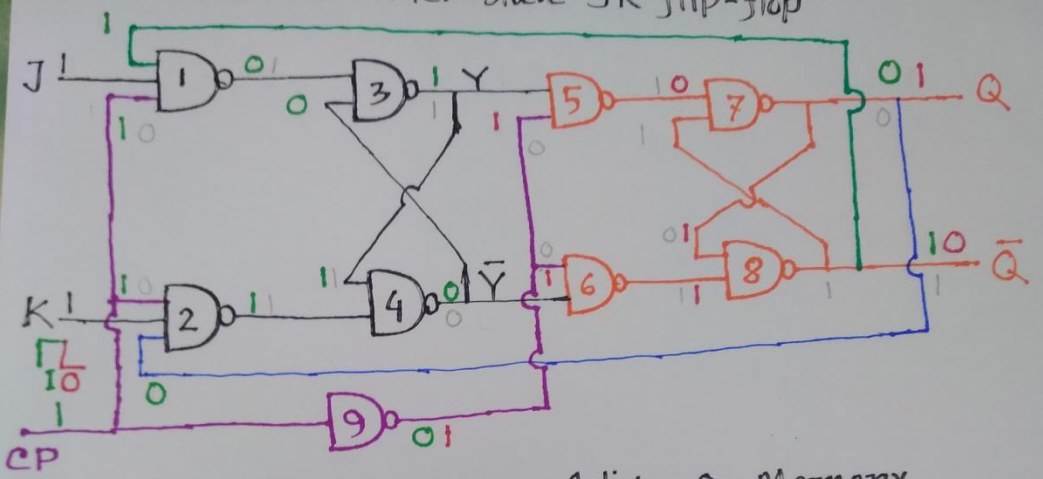


Q(t)	Q(t+1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

$$Q(t+1) = \bar{K}\,Q(t) + J\,\overline{Q(t)}$$

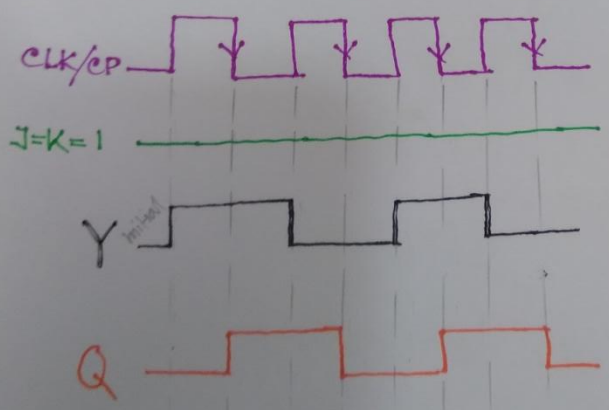


Clocked master-slave JK flip-flop



J = K = 1, CP = 1, Y = Active, Q = Memory

J = K = 1, CP = 0, Y = Memory, Q = Active



NAND

Both input 1 i.e, output 0

Any " 0 i.e, " 1

		AND	NAND
x	y	xy	$\overline{xy}$
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0